National College of Ireland

Higher Diploma in Science in Computing (HDCOMP)

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Software Development Continuous Assessment – (50%)
Individual Assignment

Release Date: Check Moodle Page
Submission Deadline:
Sunday 15 November 2026 at 23:59

IMPORTANT: PLEASE READ

- It is your responsibility to avoid plagiarism. Please read the comprehensive guidelines on academic honesty and academic integrity, and how to avoid plagiarism made available by the NCI Library: https://libguides.ncirl.ie/referencingandavoidingplagiarism.

- You are not allowed to publish this assignment brief or a part thereof on any websites. You are not allowed to publish/share your solution with others. All work submitted should be your own. Conferring with others is not permitted.

- All submissions will be electronically screened (via Turnitin) for evidence of academic misconduct (i.e. plagiarism and collusion). The examiners reserve the right to conduct live mini presentations with a sample of the students, where students will provide answers to questions related to their assignment. Also, it should be noted that an examiner may require to conduct a viva examination with you to further clarify any concerns with any submission.

SUBMISSION REQUIREMENTS

Project description

Develop a Java application for a computer repair shop. The application should allow users to enter customer details, calculate a repair quotation and view previous quotation totals.
You may use either Scanner with console output or JOptionPane for input and output.

Application menu

Display the following menu repeatedly until the user chooses to exit.

Option	Function
1	Create a quotation
2	View quotation totals and summary
3	Exit

Option 1 Create a quotation

Ask the user to enter:

- Customer name.
- One repair service from the table below. Each quotation is for one service only.
- Whether the customer is a student.
- Whether express service is required.

Service number	Repair service	Base price
1	Diagnostic check	€30
2	Software installation	€45
3	Virus removal	€60
4	Screen replacement	€120

Calculate the final quotation using these rules:

- Student customers receive a 10% discount on the base price.
- Express service adds €20 after the discount.
- No other charges apply.
- Display the customer name, repair service, base price, discount, express charge and final total. Show monetary amounts to two decimal places.
- Display €0.00 for the discount or express charge when it does not apply.

For example, a student requesting express virus removal pays €60 − €6 + €20 = €74.

- Once all inputs are valid, calculate and display the quotation and automatically store its final total in a one-dimensional double array with space for five quotations. Each completed quotation occupies one array position. Use a counter to track how many quotations have been stored. Data only needs to be retained while the application is running.
- Once the array is full, display a message and prevent additional quotations. The user must still be able to select Options 2 and 3.

Option 2 View quotation totals and summary
Use a loop to display each stored quotation total.
Also display:

- The number of quotations.
- The combined value of the quotations.
- The highest quotation.
- The average quotation value.

- Calculate the combined value, highest quotation and average by processing the stored values using a loop.
- Only process array entries containing quotations. If none have been created, display a suitable message without calculating an average.
- You do not need to store customer names or other details in arrays.

Option 3 Exit
Display a closing message and end the application.

Sample input and expected output

Action	Input	Expected output
Option 1: First quotation	Name: Alex Murphy; Service: 3; Student: Yes; Express: Yes	Customer: Alex Murphy; Service: Virus removal; Base price: €60.00; Discount: €6.00; Express charge: €20.00; Final total: €74.00
Option 1: Second quotation	Name: Sam Kelly; Service: 1; Student: No; Express: No	Customer: Sam Kelly; Service: Diagnostic check; Base price: €30.00; Discount: €0.00; Express charge: €0.00; Final total: €30.00
Option 2: View summary	Select 2 after creating both quotations	Quotation 1: €74.00; Quotation 2: €30.00; Number: 2; Combined value: €104.00; Highest: €74.00; Average: €52.00

Programming requirements

- Use service numbers 1-4 to select a repair. Accept yes/no answers regardless of capitalisation. If input is invalid, display an error message and ask for that value again. Invalid input must not create or store a quotation.
- Create an instantiable class with private attributes, a constructor, getters, setters and a method to calculate the quotation.
- Create a separate application class containing the main method.
- Validate menu and service choices, ensure the customer's name is not blank, and accept only yes/no answers for student status and express service.
- Use the programming concepts covered in the module.
- If a user enters non-numeric input or a number outside the permitted range for a menu or service choice, display an error message and ask for that value again.

Deliverables

Submit the following through the relevant Moodle submission pages:

1. A ZIP file containing all Java source files (.java).
2. A Word or PDF report containing:
   - Your name and student ID.
   - An IPO chart.
   - A brief explanation of your classes and main functionality.
   - Test inputs, expected results and screenshots showing the application working.
   - Your complete source code as text in an appendix.

Test all menu options, pricing rules, summary calculations, invalid choices and the five-quotation limit.

Marking scheme

Criteria	Marks
Instantiable class: attributes, constructor, getters and setters	20
Processing method: correct quotation calculations	20
Application class: object creation, method calls, input and output	15
Selection, menu loop and validation	10
Array and summary: storing totals, displaying quotations and calculating combined value, highest and average	10
Compiling and running application	10
Code quality: naming, indentation and useful comments	5
Report: IPO chart, explanation and testing evidence	10
Total	**100**

Scanner or JOptionPane solutions will be assessed using the same marking criteria.